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Bachelor's Thesis Template - Title of Project

Bachelor's Thesis

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Abstract: Replay-based continual learning is usually judged by the accuracy it reaches *after* each task. Continual evaluation has exposed a transient failure that this end-of-task view hides: the *stability gap*, a sharp drop in past-task accuracy at the start of every new task, before the model recovers. The gap persists even when the optimiser has access to the exact joint loss over all tasks, so it is not a defect of the objective but of the optimisation trajectory. We argue that, in its dominant components, the gap is a symptom of a single structural feature of online continual learning: at each task boundary the loss the optimiser follows changes discontinuously, from the replay loss alone to the sum of replay and current-task losses. We decompose the gap into three additive contributors — a first-step magnitude asymmetry, a finite-buffer estimator variance, and a trajectory effect — and identify the discontinuity as the source of the first and third. To remove it, we introduce the λ-*curriculum*, a homotopy that ramps the current-task loss weight from 0 to 1 over the first steps of each task, so the optimum moves along a continuous path rather than teleporting across the landscape. An envelope-theorem argument shows the replay loss is monotone non-decreasing along this path, bounding the trajectory contribution by $O(1/N)$ in the ramp length N. We add two refinements, a $\lambda_{\min}$ floor that restores early-task plasticity and momentum that suppresses the residual stochastic oscillation, and evaluate the method on rotated MNIST against vanilla and gradient-balanced Experience Replay.

1 Introduction

Deep neural networks learn rich representations from static, independent, and identically distributed data, but a vulnerability appears as soon as training becomes sequential: the network overwrites previously acquired knowledge to accommodate new data. This phenomenon, catastrophic forgetting [18, 21], is a central obstacle in machine learning: retraining from scratch on all data whenever the distribution shifts is the obvious remedy, but it is too expensive for systems that must adapt continually. Continual learning (CL) studies how to learn from a sequence of tasks without that cost, and its core difficulty is the stability–plasticity dilemma: the model must stay plastic enough to learn new distributions while staying stable enough to preserve old ones [19].

Several families of techniques address forgetting, including replay, parameter and functional regularisation, optimisation-based methods, and template-based classification [23]. These methods reduce long-term forgetting, but they have historically been evaluated only at task boundaries or at coarse intervals, which hides what happens within a task. Evaluating the network continuously exposes a failure these protocols miss. Lange et al. [16] did exactly this and identified the *Stability Gap*, a transient but severe collapse in past-task accuracy that occurs immediately upon the introduction of a new task. Even a well-tuned Experience Replay model that finishes a task with high accuracy on earlier tasks passes through a deep minimum during the first hundred steps of the new task.

This matters for three reasons. First, deployed continual-learning systems are queried throughout training, so a model mid-transition commits errors its final checkpoint never would. Second, the transience is a scientific puzzle: the model clearly retains the knowledge needed to recover, so some mechanism must temporarily disrupt and then restore it. Third, relearning what was already known wastes compute, and understanding that mechanism is a prerequisite for fixing the gap efficiently.

The discovery of the gap raised an immediate question: could it be removed by better approximating the joint loss over all tasks? Hess et al. [11] showed that the gap persists even under incremental joint training, where the model has access to all data from all tasks at every step. A perfect objective still produces transient forgetting, so the problem lies not in *what* is optimised but in *how* the optimisation proceeds.

To see why the trajectory itself produces the gap, consider the geometry of sequential optimisation. Figure 1.1 visualises a transition from a learned task (Task A) to a new task (Task B), a conceptual two-dimensional reduction of a high-dimensional surface that nonetheless captures the dynamics at play. The network starts at the converged minimum A of Task A and must reach the joint minimum $A \cap B$. Running stochastic gradient descent (SGD) on the joint loss from A follows the green curve: pulled by the steepest gradients of the joint landscape, the trajectory bows outward, leaving Task A's basin and crossing a region of higher Task-A error before converging at $A \cap B$. The dip on Task A along this arc is the stability gap.

This trajectory is the consequence of a discontinuity at the task boundary. While Task A is learned the optimiser descends the replay loss alone, the red basin in Figure 1.1. The moment Task B begins, the objective switches in a single step to the sum of the replay and current-task losses, the purple joint basin. Its minimum at $A \cap B$ replaces the red one from one step to the next. The parameters do not move during this switch, yet the minimum they are chasing jumps from θ^*_A to θ^*_{joint}. We call this *landscape teleportation*: from the optimiser's point of view the surface it was descending disappears and is replaced by another whose minimum lies elsewhere.

The teleportation does more than redirect the optimiser along a curved path; it also makes the very first step after the switch lopsided. At θ^*_A the replay gradient is near zero, because the optimiser has converged on Task A, while the current-task gradient is large, because Task B is still unlearned. The summed update is therefore dominated by the new-task gradient, and SGD takes a long step out of Task A's basin before the still-negligible replay term can pull it back. This magnitude imbalance compounds the trajectory effect: not only does the descent path bow away from Task A's minimum, the first step along it is the largest and the least opposed, which is why the dip is steepest in the first updates after a task switch [16]. A finite replay buffer adds a third, smaller error: it estimates the past-task gradient from a handful of stored samples rather than from all past data, so even the replay signal it supplies is noisy. Trajectory effect, magnitude asymmetry, and estimator variance are the three contributors to the gap; Section 3 makes their decomposition precise and the experiments measure each in isolation.

Existing methods take this discontinuity as given. They accept that the objective jumps at the task boundary and instead make the optimiser robust to the jump, constraining each SGD update with information about the past tasks, their gradients or local curvature, so that SGD avoids directions in which the replay loss rises sharply [4, 13, 14, 17]. We group these approaches under the term *path-finding*: effective as they are, they navigate the broken landscape without removing the break.

This thesis takes the opposite route. Rather than steer the optimiser carefully across a discontinuity, we propose removing the discontinuity itself, an approach we call *landscape-shaping*. We instantiate it as the **λ-curriculum**: instead of switching the objective from the replay loss to the joint loss in a single step, the curriculum ramps the current-task loss weight from 0 to 1 over the first N steps of each transition,

$$
\begin{aligned}
L_\lambda(t) &= L_{\text{replay}} + \lambda(t)\, L_{\text{current}}, \\
\lambda(t) &= \operatorname{clip}\!\left(\tfrac{t}{N}, 0, 1\right),
\end{aligned}
\tag{1.1}
$$

so the optimum $\Theta^*(\lambda)$ traces a continuous curve from θ^*_A (at $\lambda = 0$) to θ^*_{joint} (at $\lambda = 1$), and the model follows this moving valley instead of being teleported across the discontinuity (fourth panel of Figure 1.1). That such a continuous low-loss route from θ^*_A to θ^*_{joint} exists at all is what work on linear mode connectivity [6, 8] establishes: distinct optima are often joined by simple paths of near-constant loss — the dashed corridor in Figure 1.1 — and the curriculum's role is to keep the optimiser on that route rather than let it bow away. Because the optimum now moves continuously, the curriculum addresses two of the three contributors at once. It removes the trajectory effect, since the descent no longer bows away from Task A's basin (the green curve), and it tempers the magnitude imbalance, since the large new-task gradient is introduced gradually rather than dominating the first step. Only the third contributor, the variance of the

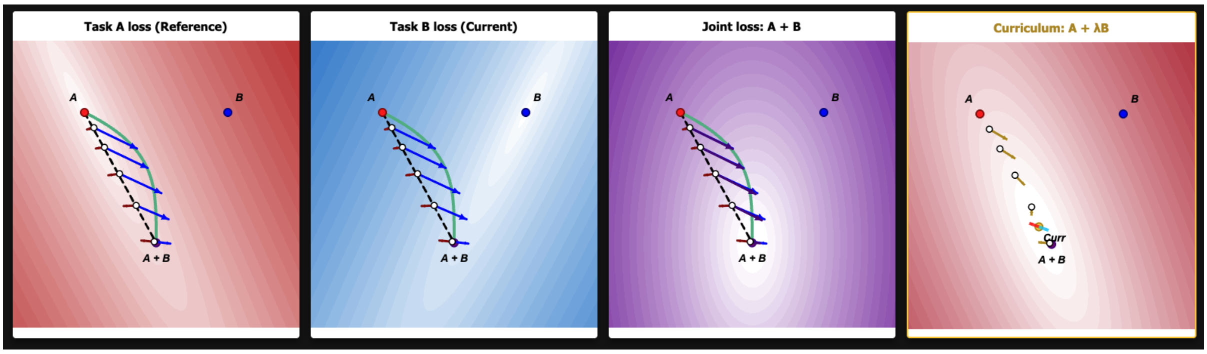


Figure 1.1: Conceptual visualisation of the optimisation landscape at a task transition, shown over a shared two-dimensional parameter space. Task A (red) and Task B (blue) are the single-task losses with minima at A and B; the joint loss $A+B$ (purple) has its minimum at $A \cap B$. The dashed line is the straight interpolation from A to $A \cap B$; the green curve is the steepest-descent trajectory of the joint loss, which bows away from Task A's basin and induces the stability gap. The fourth panel shows the λ-curriculum loss $A + \lambda B$: a small weight λ on Task B reshapes the basin into a valley that runs continuously from A toward the joint minimum, so the optimum moves along the corridor instead of teleporting across it. Arrows are normalised to a fixed length and show gradient direction only.

finite-buffer estimator, is independent of the schedule and left untouched. Treating the task-boundary discontinuity as the object to be fixed is the central contribution of this thesis; Section 4 makes the argument precise.

Two refinements extend the basic curriculum. The linear ramp requires choosing the length N; an **adaptive schedule** removes this knob by setting $\lambda(t)$ from the live ratio of replay to current-task gradient magnitudes. Because that ratio can hold λ near zero for the first steps and stall progress on the new task, a $\lambda_{\min}$ **floor** keeps the current-task gradient present from step one without re-introducing the depth the curriculum removes. Independently, **momentum** acts as a low-pass filter on the per-step accuracy oscillation that the continuous-limit argument abstracts away, suppressing mini-batch noise while preserving the slow drift along $\Theta^*(\lambda)$.

The remainder of this thesis develops these ideas. Section 2 reviews continual learning, Experience Replay, the path-finding methods we compare against, the stability-gap literature, and two lines of work — blurry task boundaries and mode connectivity — that bear on the curriculum. Section 3 then sets out the thesis's own account of the gap: it decomposes the gap into three contributors and contrasts path-finding with landscape-shaping. Section 4 proves that a continuous linear λ-ramp removes the trajectory contributor, with a residual of order $1/N$ in the ramp length. Section 5 states the research questions, and Section 6 describes the experimental testbed and the conditions used to test each prediction.

2 Background and Related Work

2.1 Continual Learning

The introduction described catastrophic forgetting and the stability–plasticity dilemma; here we place the methods that address them. Approaches to forgetting fall into three families [23]. Regularisation-based methods add a penalty that slows changes to parameters identified as important for past tasks, where importance is estimated from a diagonal approximation of the loss curvature [14]. Architecture-based methods give each task its own parameters, avoiding interference at the cost of a model that grows with the number of tasks. Replay-based methods store a small buffer of past examples and interleave them with new-task data during training. This thesis studies the replay family, and in particular its simplest member, Experience Replay.

2.2 Experience Replay

Experience Replay (ER) is the simplest replay baseline [5]. At each step it draws a replay mini-batch from the buffer, combines it with the current-task mini-batch, and takes a single optimiser step on the summed loss of the two. The buffer is populated by reservoir sampling [24], so every sample seen so far occupies a slot with equal probability regardless of arrival order. During T_0 the buffer is empty by construction, so ER reduces to plain SGD on the first task, and the stability gap is measurable only from T_1 onward.

This simplicity is also why ER remains competitive. Chaudhry et al. [5] show that jointly training on the current task and a small episodic memory significantly outperforms many purpose-built continual-learning methods. Methods that regularise heavily to protect past tasks tend to give up plasticity, whereas a small replay buffer scales more gracefully to longer task sequences. Consequently, ER serves as a good baseline. Because it does not explicitly manipulate or balance the magnitudes and directions of the gradients, it provides a clean setting to isolate the underlying causes of the stability gap. We therefore adopt it as one of our two core baselines.

A finite buffer is an imperfect estimator of the true past-task distribution. Aljundi et al. [2] frame buffer selection as a constraint-reduction problem in which the stored samples approximate the feasibility region defined by the full past data; reservoir sampling is one (non-optimal) policy within that frame. The replay gradient computed on a small buffer therefore differs in direction and magnitude from the gradient on all past data. This estimator gap is present at every step and disappears under no schedule; we report it as a separate, persistent contributor to the stability gap.

2.3 Path-Finding Methods

A second class of replay-based and replay-adjacent methods constrains the SGD update at the task boundary using information about the past tasks. They differ in what information that is. GEM stores past-task gradients and projects the new-task gradient onto the nearest direction whose inner product with every stored gradient is non-negative, so the update does not increase the loss on any past task in the buffer [17]; this uses first-order (gradient) information only. A-GEM replaces the per-task constraints with a single averaged past-task gradient, reducing the projection to one cheap operation [4]. Both act only on the current step, not across the trajectory. EWC instead uses second-order information: it adds a quadratic penalty around the past optimum weighted by the diagonal of the Fisher information, slowing changes to parameters the old task constrained [14]. NCL also draws on curvature, preconditioning the update by the precision matrix of the past-task posterior to take a natural-gradient step along directions that leave the replay loss roughly unchanged [13]. In the framing of this thesis (Section 3.2), all four are *path-finding* methods: they accept the discontinuity at the task boundary and use stored gradient or curvature information to pick a safe trajectory across it. NCL is the second baseline we compare against.

2.4 The Stability Gap

Lange et al. [16] introduced continual evaluation and used it to identify the stability gap: the transient accuracy dip on previously seen tasks at the start of new-task training. They observe that the gap is largest in the first few hundred steps of each task, that larger buffers produce smaller gaps, that recovery is partial with the minimum accuracy systematically below the pre-task baseline, and that the gap survives reservoir sampling and moderate learning rates. To explain it, they decompose the update gradient into a plasticity component that lowers the new-task loss and a stability component that preserves the old tasks, and trace the gap to the imbalance between the two at the boundary, where the new-task component dominates. Hess et al. [11] reframe this as a distinction between *what* is optimised and *how*: they show the gap persists even with a perfect approximation of the joint loss, so a better objective alone cannot remove it and the optimisation trajectory must be addressed directly. Subsequent work has shown the gap to be broad rather than incidental: it appears in joint incremental learning of homogeneous tasks [12], in continual pre-training of large language models [9], and in relation to the classification head [25], and mitigations have been proposed, from temporal ensembles [22] to explicit gap-reduction procedures [10].

2.5 The Gap Survives the Joint-Loss Oracle

The trajectory argument was made first by Kao et al. [13], who isolate a clean fact about gradient descent through a task transition and illustrate it on a low-dimensional toy problem with two task minima and their combined loss. Remove every replay confound — no buffer, no sampling noise, no magnitude imbalance — and give the optimiser exact access to the joint loss $L_{\text{joint}} = L_{T_0} + L_{T_1}$, then run SGD from a stationary point θ_0^* of L_{T_0}. This is the strongest possible replay regime, equivalent to multi-task training with perfect knowledge of the past. Even so, accuracy on T_0 dips before recovering. At θ_0^* the past-task gradient is near zero while the new-task gradient is not, so the first step moves essentially along $-g_1$, where $g_1 := \nabla L_{T_1}(\theta_0^*)$. Writing $H_0 := \nabla^2 L_{T_0}(\theta_0^*)$ for the old-task Hessian, a second-order expansion of the old-task loss gives

$$L_{T_0}(\theta_1) - L_{T_0}(\theta_0^*) \approx \tfrac{1}{2}\,\eta^2\, g_1^\top H_0\, g_1 > 0, \tag{2.1}$$

whenever H_0 has positive curvature in the direction of g_1, which it generically does at a strict minimum. The dip is not caused by bad gradients — the gradient is exact — but by the *trajectory* the optimiser takes. The conceptual figure in our introduction (Figure 1.1) is drawn in the same spirit; its green curve is precisely this steepest-descent trajectory of the joint loss. This trajectory effect is one of three contributors to the gap, which Section 3 collects into a single decomposition.

2.6 Blurry Task Boundaries

A parallel line of work relaxes the assumption that tasks arrive at sharp boundaries. Standard class- or task-incremental benchmarks present each task as a clean block: all of T_0, then all of T_1, with an instantaneous switch between them. Several settings replace this with a gradual change. In the *task-free* setting [1] the learner is never told that a boundary has occurred and the data distribution drifts continuously, so there is no privileged moment at which to consolidate. In the *blurry* setting introduced with Rainbow Memory [3] the classes of neighbouring tasks overlap in time: samples of an upcoming class begin to appear before the previous task has ended, so each task shades into the next rather than replacing it. Online formulations push this further, studying blurry, boundary-free streams at the level of individual samples under anytime-inference constraints [15]. Common to all of them is that the transition from one task to the next is spread over many steps instead of one.

This gradual transition is what connects the setting to our intervention, and it makes the blurry-boundary line the closest existing work to the λ-curriculum. Consider a step on a mini-batch that is an α-fraction T_1 and a $(1-\alpha)$-fraction T_0, with $\alpha(t)$ rising from 0 to 1 over the transition window. Its expected gradient is $\alpha\,\nabla L_{T_1}(\theta) + (1-\alpha)\,\nabla L_{T_0}(\theta)$, a weighted sum of the two task gradients. Substituting $\lambda = \alpha/(1-\alpha)$ shows this is, up to an overall scale, the gradient of $L_{T_0} + \lambda L_{T_1}$, the loss the λ-curriculum descends. A blurry data stream and the λ-curriculum therefore apply the same fix to the one-sided first step from opposite sides: the former mixes the two task gradients through the *data*, the latter through the *loss weight*. Both keep the past-task gradient present from step one while introducing the new-task gradient gradually.

The two differ in scope. Blurry boundaries achieve the mixing through the data stream and need no buffer, but once $\alpha = 1$ the old-task data, and with it the only retention pressure, is gone; the λ-curriculum is built on replay and retains an explicit replay signal at $\lambda = 1$. The curriculum is also the cleaner controlled experiment for the stability gap: it changes a single weight in the loss while data sampling, buffer contents, and replay magnitude are held fixed, which isolates the gradual ramp of the new-task gradient as the ingredient responsible for any reduction in the gap. The blurry-boundary results are consistent with ours — softer transitions forget less — but, because they vary the data distribution and the loss composition together, they do not separate which of the two is doing the work.

2.7 Linear Mode Connectivity

The introduction claimed that a low-loss corridor connects the single-task solution to the joint optimum. Work on *mode connectivity* supports this. Garipov et al. [8] and Draxler et al. [6] showed that independently trained minima of a deep network, though separated by a barrier along the straight line between them, are joined by curved paths of near-constant low loss. Frankle et al. [7] sharpened this to *linear* mode connectivity: networks that share a short period of early training can be joined by a straight low-loss path. For continual learning specifically, Mirzadeh et al. [20] found that the multitask solution and the sequentially trained solution often lie in the same basin, linearly connected, when the training regime is stable. This matters for the stability gap in two ways. It means the dip is not forced by the geometry, because a low-loss route from θ_A^* to θ_{joint}^* exists; and it sharpens what an intervention has to do, namely keep the optimiser on that route rather than let it bow away, as the trajectory argument predicts it otherwise will.

Taken together, these results circle the same object from different sides — the trajectory dip is real and survives an exact objective, softer transitions forget less, and a low-loss corridor to the joint optimum exists — without naming the mechanism that unifies them. The next section supplies it.

3 A Discontinuity Account of the Stability Gap

The previous section surveyed three findings that each touch a different face of the same object: the gap survives the joint-loss oracle (Section 2.5), gradual transitions forget less (Section 2.6), and a low-loss corridor demonstrably connects the single-task solution to the joint optimum (Section 2.7). None of that prior work names a single mechanism that ties them together. This section sets out the account this thesis defends: the stability gap is, in its dominant components, the downstream symptom of the discontinuity at the task boundary introduced in Section 1. We make the claim precise by decomposing the gap into three additive contributors (Section 3.1) and then by contrasting the two ways one can intervene on the discontinuity (Section 3.2).

3.1 Decomposing the Stability Gap

Previous work has identified various causes for the stability gap: the gradient imbalance (Section 2.4), the finite-buffer estimator (Section 2.2), and the trajectory effect (Section 2.5). Here, we unify these into a single decomposition, breaking the gap down into three distinct forces that act at different stages of training. We will isolate and measure each empirically in Section 6.

Consider the first step of vanilla Experience Replay (ER) at the start of a new task, T_1. Because the optimal parameters for the previous task (θ_0^*) form a stationary point for the replay loss, the true replay gradient at this point is near zero. Consequently, the first parameter update is driven almost entirely by the new task:

$$\theta_1 \approx \theta_0^* - \eta\,(\varepsilon(\theta_0^*) + g_{\text{new}}(\theta_0^*))\,, \qquad (3.1)$$

where $g_{\text{new}} := \nabla L_{\text{current}}$ and ε is the zero-mean error of the finite-buffer estimator. This initial step triggers three distinct phenomena, each contributing to the subsequent performance dip on T_0:

Magnitude asymmetry (G_{mag}). At θ_0^* the new-task gradient g_{new} does not vanish — the new task is still unlearned — whereas the buffer sampling noise ε is comparatively small and shrinks as the buffer grows. The first step is therefore dominated by the new task, $-\eta\, g_{\text{new}}$, and carries the iterate out of the replay basin. This is the same mechanism as the trajectory dip of Section 2.5: the second-order expansion there (Equation (2.1)) shows that any displacement away from θ_0^* raises the replay loss, so the dip appears at once — only here it is driven by the unbalanced first step rather than an exact joint step. Because the imbalance is fixed by the relative gradient magnitudes at step one, this is the dominant contributor to the gap, and corresponds to the gradient-imbalance effect observed by Lange et al. [16].

Estimator variance (G_{est}). In any individual training run, $\varepsilon \neq 0$, which adds a stochastic kick to θ_1. As discussed in Section 2.2, the replay buffer provides only a finite-sample estimate of the past data [2]. This per-step sampling noise creates a baseline loss degradation that persists regardless of the learning schedule, as it is independent of how the loss terms are composed.

Trajectory effect (G_{traj}). Even in an idealized scenario where magnitude asymmetry is corrected and buffer noise is eliminated ($\varepsilon \equiv 0$), a geometric penalty remains. As outlined in Section 2.5, an instantaneous switch in objectives forces the iterate off the optimal path of the joint loss. Along this new trajectory, the replay loss is strictly higher than it would be at the joint-loss optimum θ^*_{joint}. This is the structural contributor addressed by the envelope-theorem argument in Section 4.

Crucially, these three components dominate at different timescales: G_{mag} drives the massive disruption at step one, G_{traj} dictates the curve over the first equilibration window, and G_{est} provides a constant hum of noise across all steps.

3.2 Path-Finding versus Landscape-Shaping

The decomposition suggests two ways to intervene, distinguished by what they do with the discontinuity rather than by what curvature information they use. *Path-finding* accepts the discontinuity. The objective still jumps from L_{replay} to $L_{\text{replay}} + L_{\text{current}}$ at the boundary, and the intervention reshapes *how* the optimiser moves, replacing plain SGD with a preconditioned step $\theta_{t+1} = \theta_t - \eta\, M(\theta_t)\, \hat{g}_t$. Here M is built from past-task information — the inverse Fisher of EWC [14], the precision matrix of NCL [13], or the feasible-set projection of GEM [17]. The path-finding methods of Section 2.3 all sit here.

Landscape-shaping removes the discontinuity. It modifies the objective in a neighbourhood of the boundary so the surface the optimiser descends is continuous through the transition, leaving the geometry of the step to SGD's natural step-size adaptation. The λ-curriculum is the canonical example: the gradient followed is $\hat{g}_t = \nabla\big(L_{\text{replay}} + \lambda(t)\, L_{\text{current}}\big)(\theta_t)$ with $\lambda(t)$ a continuous schedule from 0 to 1. Two consequences distinguish it from path-finding. A preconditioner that normalises the gradient breaks SGD's step-size adaptation, whereas $\lambda(t)$ multiplies the loss and flows through the backward pass to every parameter without rescaling the step; and the preconditioned step needs a per-step matrix–vector product, whereas the curriculum adds one scalar multiply. Section 4 shows that this continuity is enough to remove the trajectory contributor G_{traj}, and that it simultaneously tempers G_{mag} by introducing g_{new} gradually rather than all at once.

4 The Envelope-Theorem Bound: Why a Linear λ-Ramp Works

When a model switches instantly from an old task to a new one (λ jumps from 0 to 1), the sudden change causes the optimizer to "cut the corner" and veer far off the ideal learning path. This violent detour causes a massive, temporary spike in forgetting, which Section 3.1 identified as the trajectory gap (G_{traj}).

This section outlines why replacing that sudden jump with a slow, continuous transition (a linear ramp over N steps) mathematically eliminates this spike. The full, step-by-step mathematical derivation of this proof is provided in Appendix **??**.

The core argument is built on two intuitive pillars:

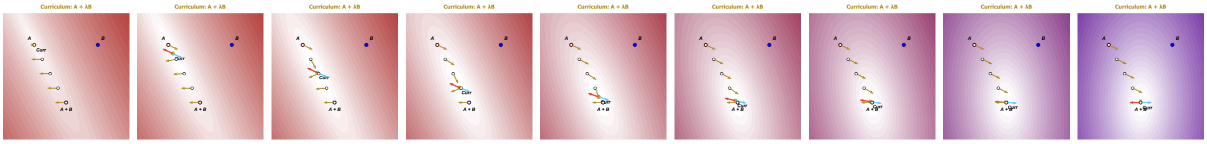
4.1 Part A: The perfect path never overshoots

Imagine an "optimum path" ($\Theta^*(\lambda)$) that perfectly tracks the exact bottom of the combined loss valley as the curriculum transitions from pure replay ($\lambda = 0$) to the joint task ($\lambda = 1$). Figure 4.1 shows this path directly: as λ grows, the minimiser of the curriculum loss $L_\lambda = A + \lambda B$ migrates smoothly from the old-task optimum A to the joint optimum $A+B$, so a model that follows it rides a moving valley floor rather than jumping between the two.

If the model were capable of perfectly walking this ideal path, the replay loss would never experience a sudden spike. As proven in the Appendix using the envelope theorem, the rate of change of the replay loss along this path is mathematically guaranteed to be non-negative:

$$\frac{d}{d\lambda} L_{\text{replay}}(\Theta^*(\lambda)) \geq 0 \qquad \text{for all } \lambda \in [0, 1]. \quad (4.1)$$

Because the replay loss can only strictly rise (or stay flat) as it accommodates the new task, it can never decrease. Therefore, it is physically impossible for the loss to spike up and come back down. It simply rises monotonically until it reaches its permanent, unavoidable baseline at the joint optimum (θ^*_{joint}).



0 0.005 0.025 0.07 0.175 0.35 0.499 0.749 1

Figure 4.1: Evolution of the curriculum loss $L_\lambda = A + \lambda B$ over a shared two-dimensional parameter space. Left to right, the weight on the new task ramps from $\lambda = 0$ (pure replay) to $\lambda = 1$ (the joint loss); the value of λ is printed beneath each panel. A and B mark the two single-task minima and $A{+}B$ the joint minimum. As λ increases the contours morph continuously from the old-task basin around A to the joint basin around $A{+}B$, and the current minimiser (*Curr*) slides along the gold chain from A toward $A{+}B$. That chain is the optimum path $\Theta^*(\lambda)$ of Section 4.1. The opposing red and blue arrows at the current minimiser are the replay- and current-task gradient contributions, which cancel there by the first-order condition $\nabla L_{\text{replay}} + \lambda \nabla L_{\text{current}} = 0$. Because the minimiser never jumps — it only drifts as λ grows — a model that tracks it rides the moving valley floor instead of being teleported across the landscape.

4.2 Part B: A slow ramp keeps the model on the path

Part A describes a mathematically perfect path, but real SGD iterates ($\theta(t)$) are clumsy and take discrete steps. However, by analyzing the gradient flow, we can prove that the actual model trajectory acts as though it is tethered to the perfect path.

The distance the model lags behind the perfect path is directly proportional to how fast the valley floor is moving. If the transition is stretched over a sufficiently long ramp (large N), the maximum distance the model wanders off the ideal path is tightly bounded:

$$\|\theta(t) - \Theta^*(\lambda(t))\| = O\left(\frac{1}{N}\right). \tag{4.2}$$

4.3 The Combined Bound

By combining the "no-overshoot" rule of the perfect path (Part A) with the tracking error of the actual network (Part B), we arrive at the final bound for the highest replay loss the model will experience during the transition:

$$\max_t L_{\text{replay}}(\theta(t)) \;\le\; L_{\text{replay}}(\theta^*_{\text{joint}}) + O\left(\frac{1}{N}\right). \tag{4.3}$$

The first term is the genuine plasticity–stability trade-off: the permanent capacity cost of learning both tasks. The second term is the transient stability gap (G_{traj}).

Because the transient gap shrinks proportionally with $1/N$, a sufficiently slow continuous curriculum physically drives the trajectory overshoot to zero. The model's step-size adaptation is left untouched, cleanly resolving the gap without the need for complex path-finding interventions.

5 Research Questions

The decomposition of Section 3.1 and the bound of Section 4 make a set of falsifiable predictions about how the λ-curriculum and its refinements should behave. We evaluate them on rotated MNIST with a small multilayer perceptron, the regime in which the envelope-theorem bound is sharply testable, using vanilla and gradient-balanced Experience Replay as baselines. Five research questions structure the experiments.

RQ1 (linear curriculum). Does the linear λ-curriculum applied to standard ER monotonically reduce stability-gap depth as the ramp length N grows, and does it do so without the final-accuracy cost incurred by gradient balancing?

RQ2 (adaptive curriculum). Does an adaptive λ-schedule, which sets λ from the live ratio of replay-to-current gradient magnitudes and so removes the explicit knob N, reduce gap depth without that final-accuracy cost?

RQ3 (momentum). Does adding momentum on top of the curriculum reduce the residual depth fluctuation predicted by the discrete-time stochastic argument? And does momentum on its own, without the curriculum, produce any comparable benefit — the prediction being that it does not, since the one-sided first step is deterministic?

RQ4 ($\lambda_{\min}$). Does a small floor $\lambda_{\min} > 0$ on the adaptive schedule recover early-task learning speed without re-introducing the depth pathology the curriculum was designed to remove?

RQ5 (decomposition). How does the gap depth decompose into the magnitude, estimator, and trajectory contributions of Section 3.1, and what does each share reveal about why a curriculum is the right shape of fix?

Section 6 describes the testbed, the metrics used to measure gap depth and final accuracy, and the full set of conditions that isolate each contributor and test each prediction.

6 Methods

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